Scale Is Not Symmetric: Directional Scale Reliability in Chart Question Answering

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Abstract

Chart question answering models must preserve the numerical scale expressed by axes, labels, and units. We study a sparse but consequential failure in which a model predicts a numerically related answer at the wrong scale, such as *0.61* instead of *61*. Our central finding is that this cannot be repaired by unconditional numeric normalization or a one-direction correction: scale correction is *directional* and *regime-dependent*. We construct ScaleQA-Natural, a mined and author-adjudicated diagnostic set of 520 natural scale-conflict candidates (473 image-evaluable), and add a blinded, construction-independent verification of strict scale errors on the ChartQA test set. Using directional scale-contrastive LoRA adapters, prompting, verifiers, calculators, and selectors as probes on an answer-tuned Qwen2.5-VL-7B, we find that (i) a prefer-large adapter recovers most verified strict test-set scale errors (14/17) and gives a small but significant held-out gain, whereas compute-matched continuation controls recover almost none (1/17); (ii) the same adapter is unsafe as a global policy, because the correct direction reverses across subtypes; and (iii) models often already generate the correctly scaled answer (blind pass@20 of 79.3% versus 52.4% single-sample), while label-free selectors recover only a small part of that headroom. Our strongest evidence concerns percentage–decimal, factor-of-100 errors on ChartQA-like data; we scope our claims accordingly and argue for regime-aware scale-reliability evaluation with fix–break accounting.

1 Introduction

The same chart value can be verbalized as a percentage, a decimal, a raw value, or a value in thousands or millions. A chart question answering model must therefore do more than find the relevant bar or label: it must decide which scale the answer should use.

Modern multimodal large language models (MLLMs) achieve strong aggregate accuracy on chart QA benchmarks such as ChartQA (Masry et al., 2022). Aggregate accuracy can nevertheless hide rare failures that are costly when they occur. We study *scale inconsistency*: a model gives a numerically related answer under the wrong scale, such as predicting *0.61* when the chart question requires *61*. This is not a generic numeric miss. The model has often found the right quantity but has failed at the last reliability step: choosing the scale in which to express it. A plausible-looking answer that is off by a factor of 100 is more dangerous than an obviously wrong one.

The unconditional interventions we test do not solve this. Normalizing percentages, training an adapter that prefers larger numbers, or asking the model to self-check units each helps some cases and hurts others: a legitimate *0.0593* should not be expanded, and a prediction that is already too large needs the opposite intervention from one that is too small. The central question is therefore not whether a model can *produce* a scaled alternative, but whether a system can *select* the correct scale regime: the direction, unit context, and output convention required by the chart and question.

Natural strict scale errors are sparse under any single model and split, so the evidence must be read at two levels. Across the ChartQA validation/test outputs we audit, strict factor-of-100 candidates account for 0.12–1.09% of examples (Table 10). We treat the automatic ratio rule as a *candidate filter*, not as proof of a scale error, and verify test-set candidates with a blinded protocol in which the annotator sees only the chart,

question, gold answer, and prediction (§3). To understand the mechanism behind these sparse failures, we also construct ScaleQA-Natural, a mined and author-adjudicated set of 520 natural scale-conflict candidates, 473 of which are directly image-evaluable (ScaleQA-Natural-473).

Our main empirical finding is that scale correction is directional and regime-dependent. In an answer-tuned Qwen2.5-VL-7B setting, a prefer-large scale-contrastive adapter improves held-out ChartQA test accuracy by +0.72 points (paired bootstrap 95% CI $[+0.38, +1.09]$) and recovers 14/17 verified strict test-set scale errors; compute-matched continuation controls with ordinary answers, random numeric negatives, or non-directional pairs recover at most 1/17, so the effect is attributable to directional scale supervision rather than extra training. The aggregate gain is small because the failure is rare; the targeted effect is large. Yet the same adapter is unsafe when applied globally: on ScaleQA-Natural it breaks more examples than it fixes, and the best direction reverses across subtypes. A blind-generation analysis shows the model often already produces the correctly scaled answer (pass@20 79.3% versus 52.4% single-sample), while label-free selectors recover only a small part of that headroom. Candidate generation and regime selection both matter, and regime selection remains the larger open gap.

Contributions. We contribute (i) a verified strict scale-error recovery probe on ChartQA test and ScaleQA-Natural, an audited diagnostic set of natural scale conflicts; (ii) evidence, with compute-matched controls, that directional scale supervision repairs a verified error slice but is unsafe as a global policy; (iii) a decomposition of candidate generation from regime selection, using prompts, verifiers, calculators, likelihood scoring, blind sampling, and label-free selectors; and (iv) a scoped conclusion: for percentage–decimal, factor-of-100 errors on ChartQA-like data, reliable scale-regime selection, not candidate availability, is the central unresolved problem.

2 Related Work

Chart question answering. Chart question answering requires models to combine visual perception, OCR, structural grounding, and numerical reasoning. Earlier benchmarks such as DVQA, FigureQA-style synthetic chart reasoning, and PlotQA emphasize controlled visual reasoning over chart data (Kafle et al., 2018; Methani et al., 2020). ChartQA (Masry et al., 2022) evaluates both human-written and machine-generated chart questions that involve value lookup, comparison, and arithmetic. Recent chart-specialized systems add plot-to-table reasoning, chart derendering, chart-specific pretraining, and visual instruction tuning (Liu et al., 2023a,b; Masry et al., 2023; Han et al., 2023; Masry et al., 2024). Recent benchmarks stress more complex and realistic charts (Xu et al., 2023; Wang et al., 2024; Masry et al., 2025; Wei et al., 2025), and ChartInsights (Wu et al., 2024) evaluates low-level chart tasks with fine-grained error taxonomies. These diagnostics slice errors by task type, chart type, or reasoning step; none isolates the case in which the model finds the right quantity but reports it under the wrong scale, nor tracks the *direction* of that error. ScaleQA-Natural is complementary: it is a narrow, directional slice that can be mined from any of these benchmarks, and our fix–break accounting applies to any correction evaluated on them.

Numerical reliability in MLLMs. Numerical errors in multimodal models arise from grounding failures, OCR mistakes, wrong axis interpretation, arithmetic errors, or answer-normalization artifacts. Scale inconsistency differs from a generic numeric miss because the prediction is close under a simple multiplicative transformation, which makes it easy to miss in aggregate error analysis and risky in deployment. The evaluation target is therefore not only whether a model can retrieve a number, but whether it chooses the output convention supported by the chart's axis, unit, and question wording.

Diagnostic evaluation and correction policies. Most chart QA work reports aggregate accuracy and analyzes failures post hoc. We instead treat scale conflicts as a diagnostic slice with inclusion criteria, an audit trail, and fix–break accounting, and we separate candidate generation from candidate selection, a distinction single-answer accuracy cannot cap-

ture. TextVQA (Singh et al., 2019) serves only as an adjacent text-rich safety check.

Parameter-efficient and preference-style adaptation. LoRA updates a model through low-rank trainable matrices while keeping the base weights fixed (Hu et al., 2021). This makes it attractive for targeted reliability interventions in large MLLMs. Contrastive and preference-style objectives, including direct preference optimization (Rafailov et al., 2023), optimize relative preferences between candidate outputs. Our objective is more local: for a chart and question, we contrast a preferred numeric answer with a scale-related alternative, and study how the direction of this contrast changes behavior.

3 Diagnosing Scale Inconsistency

We use three views of the problem: official ChartQA accuracy measures task utility, verified strict scale-error recovery measures whether real scale mistakes can be repaired, and ScaleQA-Natural exposes why a repair that helps one regime can damage another.

Strict scale-error candidates. Let y be the gold answer and $\hat{y}$ a model prediction. We extract a single numeric value $n(\cdot)$ when possible and flag an example as a *strict scale-error candidate* if the baseline prediction is incorrect, both answers contain one nonzero number, and their ratio is approximately 100 or 1/100:

$$\frac{|n(\hat{y})|}{|n(y)|} \in [80, 125], \quad \text{or} \quad \frac{|n(\hat{y})|}{|n(y)|} \in [0.008, 0.0125]. \tag{1}$$

A percent sign is stripped before parsing, so *0.61%* is parsed as 0.61. This rule is a *candidate filter*: a ratio near 100 can also arise from a wrong entity, a wrong operation, or a coincidental numeric miss. We therefore verify candidates manually before using them as a recovery probe.

Blinded verification on ChartQA test. To obtain evidence that is independent of the ScaleQA-Natural construction, we scanned all numeric-answer ChartQA test examples with the ratio rule and verified every strict candidate through a blinded interface that shows only the chart, question, gold answer, and model prediction, hiding the candidate group, prediction/gold ratio, and relation label. Of 18 strict factor-of-100 candidates, 17 were verified as true scale errors and 16 as clean scale errors with no confounding issue; among 30 randomly sampled non-candidates, 1 was judged a missed scale error. The verified set (17 examples) is the recovery probe used in §5. Baseline accuracy on it is zero by construction, so recovery is reported as counts with Wilson 95% intervals (items: Appendix E).

ScaleQA-Natural. Because strict errors from a single model and split are sparse, we mine scale-conflict candidates from ChartQA predictions across model outputs and split shards and review them with a machine-assisted annotation workflow. ScaleQA-Natural makes this rare failure mode measurable without turning it into a hidden training set or leaderboard. It contains 520 author-adjudicated positives, of which 473 are directly image-evaluable; we call this subset *ScaleQA-Natural-473* and use it for all image-conditioned experiments. Released items include question, gold answer, source row, model prediction, subtype, direction, review label, and audit metadata; images are referenced through original ChartQA rows. We also retain the 403-item seed set, *ScaleQA-Core*, and a 192-example InternVL-mined subset for source-distribution analysis.

Annotation pipeline and what the audit does and does not establish. The pipeline has three stages that we report separately (Table 13, appendix). (1) *First-pass labeling*: 739 mined candidates receive include/review/reject labels. (2) *Author adjudication*: a deterministic metadata audit is compared with the first-pass labels; the $\kappa = 0.443$ reported there is agreement between this automatic audit and the first-pass labels, *not* human–human agreement, and it flags 108 of the eventual positives for manual adjudication. (3) *Inclusion audit*: a second human reviewer audited the 520 final positives and made no changes. This second pass reviewed retained positives only, so it provides no estimate of false negatives in the rejected or uncer-

tain pool, and its 100% agreement should not be read as independent initial label reliability. We therefore describe ScaleQA-Natural as an author-adjudicated, inclusion-audited diagnostic set, not an independently human-authored benchmark. The blinded ChartQA-test verification above is the construction-independent check on the phenomenon itself.

Split hygiene. The main Qwen scale adapters are trained from ChartQA training rows and training-set prediction/log-probability caches. ChartQA validation is used for development; held-out ChartQA test is reserved for the main task result. ScaleQA-Natural is used for diagnosis and policy-selection analysis, not for training the reported Qwen scale adapters. Because ScaleQA-Natural is mined from model outputs on several ChartQA shards, 363 of the 473 image-evaluable items (76.7%) come from ChartQA *training* rows that the answer-tuned adapter saw during answer tuning; 110 items come from validation/test rows. [State whether any of these 363 rows are among the $K = 300$ scale-pair training rows.] We therefore report every ScaleQA-Natural adapter result on both the training-row and the row-disjoint subsets (§5.2); the direction-dependent pattern is the same on both, but the sign of the aggregate adapter effect is not. Automatically mined candidates that remain uncertain or are rejected often involve plausible ratios but insufficient evidence that prediction and gold refer to the same chart quantity; these are excluded rather than treated as positives (Table 11).

Why this decomposition matters. ChartQA test asks whether an intervention helps the task, the verified strict subset asks whether it repairs natural scale errors, and ScaleQA-Natural asks why the same intervention helps one regime but harms another. We therefore report full accuracy, strict recovery counts, non-scale accuracy, and fix–break accounting throughout.

4 Diagnostic Probes and Policy Selection

We freeze the MLLM backbone and train LoRA parameters Δ (Hu et al., 2021); for chart x, question q, and answer y the adapted model defines $p_{\theta,\Delta}(y \mid x, q)$. The methods form a probe ladder, not a single fix: directional adapters test whether scale preference can be moved, compute-matched and clean-direct controls test why and whether safely, blind sampling tests whether the correct answer is generated at all, and selectors test whether candidate evidence yields a regime decision.

Chart + question → Base answer (e.g., *0.61*)

Candidate pool: base, self-check answer, directional adapter answers, blind samples, explicit scale variants.
Regime evidence: question form, answer magnitude, percent/unit tokens, axis labels, tick values, candidate likelihoods.
Policy decision: base, self-check, prefer-large, prefer-small, decimal guard, or unit guard.

Prediction too small: 0.61 → 61
Prediction too large: 61 → 0.61
True small decimal: keep 0.0593
Unit mismatch: millions vs. raw values

Figure 1: Regime-aware diagnostic view. The experiments test whether scale-related candidates can be generated and, more importantly, whether a deployable policy can select the scale regime required by the chart and question.

Directional adapters as probes. For numeric training examples we construct scale-related pairs (y^+, y^-) in which y^- is y^+ rescaled by a factor of 100 or $1/100$ (with percent signs handled as in §3; full pair-construction rules, rounding, and sign handling are in Appendix A). *Prefer-large* pairs take the larger-scale value as y^+; *prefer-small* pairs take the smaller. Tables retain the training-time labels *large-positive*/*small-positive*, which name the same preferences. The names describe the training preference, not the observed error direction: an answer that is too small requires a prefer-large correction, and vice versa. We also evaluate balanced mixtures, all mined pairs, and real-conflict pairs mined from model disagreements.

Objective. We compute mean token negative log-likelihood $\mathrm{NLL}(y \mid x, q)$ and minimize

$$\mathcal{L}_{\text{scale}} = \mathrm{softplus}\big(-\beta[\mathrm{NLL}(y^-) - \mathrm{NLL}(y^+) - m]\big), \quad (2)$$

where m is a margin and β controls sharpness. The loss encourages the preferred answer to have lower NLL than its scale-related alternative.

Adaptation regimes and controls. The main positive result uses an *answer-tuned* starting point: scale adaptation continues from an answer-only LoRA adapter in the Qwen2.5-VL-7B pipeline for 300 additional steps. Because any continuation could help simply through more optimization, we add *compute-matched controls* that continue the same answer-tuned adapter for the same 300 steps with (a) ordinary answer examples, (b) random or shuffled numeric negatives, and (c) non-directional balanced pairs with matched answer frequency. We also test *clean direct* scale-only tuning from the raw backbone as a safety control. At inference, main results use the same deterministic short-answer decoding template as the baseline, without reranking or calculators.

Blind-generation oracle. To test whether candidate oracles are high only because pools contain deterministic transforms of the base answer, we also draw 20 stochastic samples with no scale prompt and no transformation and report pass@20 against single-sample accuracy.

Label-free policy selection. ScaleQA-Natural suggests that no single direction is safe across all regimes. We therefore evaluate selectors that choose among base answering, self-check prompting, unit/axis guards, and directional adapters, using question text, base prediction text, numeric features, and chart-context strings extracted from the image (title, axis labels, tick values, legends, units). We call a selector *label-free* when it uses no gold subtype or direction field; this is a weaker guarantee than an untouched test set, and we report selector results under a fixed partition and leave-source-out evaluation accordingly (§6.2). Oracle subtype, policy-pool, and candidate-pool results are upper bounds whose role is to show how much headroom remains if regime detection were solved.

System	Starting point	Test acc.	Std.	Delta
Raw Qwen2.5-VL-7B	Raw backbone	0.7804	–	–
Answer-tuned LoRA	Answer-only	0.8860	–	+0.1056
Answer-tuned + large-positive scale	Answer-tuned	0.8932	0.0003	+0.0072
Clean direct large-positive scale	Raw backbone	0.7456	–	-0.0348
Clean direct small-positive scale	Raw backbone	0.7136	–	-0.0668

Table 1: Held-out ChartQA test attribution for Qwen2.5-VL-7B. The scale update is measured on top of the answer-tuned adapter; clean direct scale-only tuning is a safety control. "Large-positive" denotes the prefer-large adapter.

5 Experiments

Setup. The main experiments use Qwen2.5-VL-7B-Instruct (Qwen Team, 2025). We use ChartQA validation (1,920 examples) for development and held-out ChartQA test (2,500 examples) for the main task result (Masry et al., 2022). We also evaluate InternVL2.5-1B/8B/26B (OpenGVLab, 2024) and numeric subsets of PlotQA (Methani et al., 2020), ChartBench (Xu et al., 2023), MathVista (Lu et al., 2023), and InfographicVQA (Mathew et al., 2022). Numeric answers are scored with the repository evaluator: normalized exact match for text answers and 5% tolerance for numeric answers. The final Qwen scale adapters use $K = 300$ pairs, 300 steps, learning rate 10^{-6}, $\beta = 2.0$, and margin $m = 0.10$. All accuracy differences are in percentage points.

5.1 Held-Out ChartQA and Verified Strict Errors

The first question is deliberately narrow: can a scale-directed update fix natural scale errors without damaging the main task, and is the effect due to the scale objective rather than extra training?

Table 14 separates answer tuning from scale tuning. The raw pipeline obtains 78.04 on ChartQA test; answer tuning reaches 88.60. Continuing from that adapter, prefer-large scale-contrastive tuning reaches 89.32 over five seeds, +0.72 points with paired bootstrap 95% CI $[+0.38, +1.09]$. Clean scale-only adaptation from the raw backbone hurts test accuracy by 3.5–6.7 points. The aggregate gain is small because strict scale errors are rare (Table 10); non-scale accuracy moves only from 89.35 to 89.66 (Table 22), and TextVQA-500 is unchanged at 94.20 (a small adjacent-task check, not a general safety guarantee).

Continuation from answer-tuned adapter (300 steps, 3 seeds)	ChartQA val. acc.	Verified strict test errors recovered
None (answer-tuned baseline)	0.8640	0/17 [0.000, 0.184]
Ordinary answer replay	≈0.8640	1/17 [0.010, 0.270]
Random / shuffled numeric negatives	≈0.8640	[fill]
Non-directional balanced pairs	≈0.8640	[fill]
Directional scale pairs (prefer-large)	0.8695	12/17 [0.469, 0.867]
Prefer-large adapter, main setting (5 seeds)	–	14/17 [0.590, 0.938]

Table 2: Compute-matched controls. Every row continues the same answer-tuned adapter for the same number of steps; only the continuation data differs. Recovery is reported as counts over the 17 blind-verified strict test errors with Wilson 95% intervals.

Table 15 answers the attribution question. On the verified strict test errors the answer-tuned baseline recovers 0/17. Continuing with ordinary answer examples recovers 1/17, and non-directional or random-negative continuation is statistically tied with the baseline on ChartQA validation. Directional scale supervision recovers 12/17 in the three-seed control setting and 14/17 with the main five-seed prefer-large adapter. The intervals are wide, but they exclude the control outcomes: the targeted effect comes from directional scale supervision, not from 300 more steps. This is a verified diagnostic-slice result, not a population-level rate (per-seed counts: Appendix E).

5.2 ScaleQA-Natural

The second question is whether the same intervention is safe when scale conflicts are collected across regimes rather than filtered to one verified recovery subset.

On ScaleQA-Natural-473, answer tuning improves the raw backbone from 45.67 to 50.95, a scale-aware prompt drops to 44.19, and the global prefer-large adapter reaches 49.51, *below* the answer-tuned base. On ScaleQA-Core, balanced and all-pair variants outperform the answer-tuned base; on the InternVL-mined subset, prefer-small adaptation is far stronger than every other global policy (74.90 versus 33.23 balanced).

Table 4 separates the items by whether their ChartQA row was seen during answer tuning. The aggregate "adapter hurts" result is carried entirely by the training-row items (-4.1 points, CI $[-7.8, -0.4]$, breaking 8.8%). On the 110 row-disjoint items the same adapter *helps* ($+7.3$ points, CI $[+1.5, +13.6]$, fixing 9.1% and breaking 2.7%) and edges self-check (58.18 versus 55.45). The direction breakdown shows why: in both subsets prefer-large helps when the prediction is too small and hurts when it is already too large; the subsets differ mainly in their mix of directions. Directionality is the stable finding; the sign of a global adapter's aggregate effect on a mined set depends on the set's direction composition, itself a reason not to deploy unconditional correction.

Training-free prompting baselines (Table 32, appendix) are stronger than the global adapter on the full set. A self-check prompt reaches 54.55; majority voting adds little, while oracle prompt choice reaches 67.44. The question becomes whether the right answer is missing or merely not selected.

Table 18 addresses a construction concern directly: because ScaleQA-Natural is mined through multiplicative relations and the structured oracle pools contain explicit scale variants, oracles over those pools are high partly by construction. The blind-generation row removes both. With no scale prompt and no transformation, 20 stochastic samples already contain a correct answer for 79.3% of examples, against 52.4% for a single sample. Deterministic variants raise coverage only further, to 91.75–92.60; most of the gap to deployable methods is present without them.

5.3 Cross-Backbone and External Checks

The final question is scope: does the phenomenon extend beyond the main Qwen setting, and does the intervention transfer with it? Table 19 (appendix) summarizes the picture. InternVL2.5-1B and 8B can be adapted with the same scale-pair recipe, and InternVL2.5-26B serves as a stronger diagnostic backbone on ScaleQA-Natural, PlotQA, ChartBench, MathVista, and InfographicVQA subsets. These checks establish two narrow facts: ratio-related scale errors occur under every backbone we test, and the prefer-large adapter does not transfer as a correction (InternVL2.5-8B shows only tiny differences among directions, and Qwen external subsets are unchanged across adapters). Because models, prompts, and subset sizes differ, they do not establish a common prevalence or taxonomy across models; we use them to bound the method's scope.

Set	N	Raw	Ans.	Prompt	Large	Bal.	All	Real	Small
ScaleQA-Natural-520	473	0.4567	0.5095	0.4419	0.4951	–	–	–	–
ScaleQA-Core	403	0.4591	0.4963	0.4293	0.4844	0.5092	0.5047	0.5002	0.4174
InternVL-mined	192	0.1510	0.3281	0.0990	0.1896	0.3323	0.2771	0.2510	0.7490

Table 3: ScaleQA-Natural diagnostic results. The first row is the image-evaluable ScaleQA-Natural-473 subset; ScaleQA-Core and InternVL-mined report five-seed pair-mode means. Large/Small denote prefer-large/prefer-small adapters.

ScaleQA-Natural-473 subset	*N*	Answer-tuned	Prefer-large	Self-check	Δ(large−ans.) [95% CI]	Fix	Break
Items from ChartQA training rows	363	50.96	46.89	54.27	−4.1 [−7.8, −0.4]	4.7	8.8
Items from ChartQA val/test rows (row-disjoint)	110	50.91	58.18	55.45	+7.3 [+1.5, +13.6]	9.1	2.7
All	473	50.95	49.51	54.55	−1.4 [−4.7, +1.7]	5.7	7.4
By direction (answer-tuned / prefer-large / self-check):							
Prediction too large, training rows	205	30.7	15.3	45.4			
Prediction too small, training rows	158	77.2	87.8	65.8			
Prediction too large, row-disjoint	36	22.2	14.4	33.3			
Prediction too small, row-disjoint	74	64.9	79.5	66.2			

Table 4: Row-overlap analysis on ScaleQA-Natural-473. Prefer-large is the five-seed mean; Δ is the paired per-item difference with a bootstrap 95% CI; Fix/Break are the percentages of items the adapter corrects/breaks relative to the answer-tuned model. The direction asymmetry is the same on both subsets; the aggregate sign differs because 56% of training-row items are prediction-too-large versus 33% of row-disjoint items.

Candidate source on ScaleQA-Natural-473	Accuracy / coverage
Single greedy sample (answer-tuned)	0.524
Blind pass@20, no scale prompt, no transforms	0.793
Prompt + adapter candidates (policy-pool oracle)	0.9004
Top-8 candidates incl. deterministic scale variants	0.9175
Augmented structured oracle	0.9260

Table 5: Candidate coverage on ScaleQA-Natural-473 decomposed by candidate source. Blind sampling uses no scale prompt and no deterministic transformation, so its coverage is not implied by the mining rule.

6 Analysis

Scale interventions create useful alternatives; the system fails at deciding when to trust them.

6.1 Pair Direction Is Not Symmetric

In the development ablation (Table 20, appendix), prefer-large pairs give the best full-validation score while prefer-small pairs reduce it by 1.51 points; yet ScaleQA-Natural shows regimes where prefer-small is best. Pair direction changes both which scale conflicts are repaired and which non-scale examples are broken. On ChartQA validation, the prefer-large adapter fixes a mean of 13/23 strict candidates across seeds while breaking 1.27% of base-correct examples (Table 25).

Fix–break asymmetry on ScaleQA-Natural. On ScaleQA-Natural-473 the same adapter fixes 5.71% of examples the answer-tuned model misses but breaks 7.40% of examples it gets right, a net loss. The group diagnostic (Table 36) makes the reversal concrete: prefer-large policies are strong when the original prediction is too small but poor when it is already too large, and self-check is safer in the latter regime but leaves large headroom in the former. The structured verifier shows the same asymmetry at the example level despite seeing the same image and candidate list. This is why a global adapter can help the verified test-set slice yet fail as a deployment policy: the correct action depends on question form, unit evidence, and the direction of the original conflict.

6.2 Regime Selection Is the Bottleneck

The subtype reversal suggests that correction should be selected rather than applied globally. On ScaleQA-Core a simple subtype router reaches 50.71 against 49.97 for the best fixed policy, while the oracle-subtype router reaches 80.77 (Table 21, appendix): the missing component is regime detection, not another adapter.

Table 6 compares the strongest selector with fixed policies, verifiers, and oracles. *Protocol.* For each of 50 random 60/20/20 partitions, every selector configuration (feature

Method on ScaleQA-Natural-473	Acc.
Self-check prompt (fixed policy; 50 splits)	55.14 ± 4.30
Label-free selector (50 splits)	56.67 ± 5.61
paired Δ vs. self-check	+1.54 [+0.19, +2.88]
fixed policy chosen on dev	52.74
best fixed policy per test split (oracle)	57.14 ± 3.62
Label-free selector, leave-source-out (Table 7)	Δ −1.0 to −6.9
Pairwise top-8 verifier	54.33
Structured-context verifier	48.41
Table extraction + calculator	42.28
Policy-pool oracle	90.04
Candidate top-8 oracle	91.75

Table 6: Deployable selection versus fixed policies and oracles on ScaleQA-Natural-473. "Label-free" means no gold subtype or direction feature is used; it does not mean an untouched test set, so we additionally report leave-source-out evaluation. The full ladder of selector variants is in Appendix D.1.

Held-out mining source	N_{test}	N_{train}	Selector (10 seeds)	Self-check	Δ	Policy oracle
ChartQA train shard	316	157	45.32 ± 4.98	52.22	−6.9	83.86
ChartQA val	71	402	59.01 ± 4.22	64.79	−5.8	83.10
ChartQA train	47	426	65.53 ± 7.82	68.09	−2.6	80.85
ChartQA test	39	434	37.44 ± 5.82	38.46	−1.0	79.49

Table 7: Leave-source-out evaluation of the same selector. Each ChartQA source split that contributed ScaleQA items is held out in turn; configuration is chosen on a dev partition of the remaining sources. The selector is below self-check on every held-out source.

model, stack-vs-patch mode, switch margin) and every fixed policy is scored on the 20% dev partition only; the single best configuration is refit on train+dev and scored once on the 20% test partition. The label-free selector reaches 56.67 against split-matched self-check at 55.14; the paired per-split difference is +1.54 points (bootstrap 95% CI $[+0.19, +2.88]$; Wilcoxon $p = 0.028$; 30 wins / 5 ties / 15 losses, sign test $p = 0.036$). Two comparisons bound this result. A fixed policy chosen on the same dev partition scores only 52.74 on test, below self-check: naive dev-based policy choice overfits, and the selector's gain comes from switching conservatively (48.9% of items) rather than from strong regime detection. An oracle that picks the best fixed policy per test split reaches 57.14, so the selector does not yet beat a well-chosen fixed prompt.

The advantage does not survive source shift. Table 7 holds out each mining source in turn: the selector falls below self-check on all four, by 1.0 to 6.9 points, while the policy oracle stays at 79–84. Since selector features were also iterated over several versions on this collection (Appendix D.1), even the random-split gain is an upper estimate. The selector is therefore not a deployable method; it shows that chart-side evidence carries a weak, source-specific regime signal while most oracle headroom remains unclaimed. More aggressive selectors (statistical fusion, Bayesian priors, Bayesian-MLP) fall below self-check even on random splits (Appendix D.1).

6.3 What the Negative Results Establish

The negative results rule out simple explanations for the gap between generation and selection. Table extraction plus calculation tests whether the issue is missing arithmetic; it is not (42.28). Structured verifiers test whether exposing chart context and explicit scale variants is enough; it is not (48.41). Candidate-likelihood and pairwise-verifier variants test whether the model internally prefers the right candidate once shown a compact list; again no. Blind sampling (Table 18) shows that the missing answer is usually not missing: the model generates the correct scale in 79.3% of cases within 20 samples. Candidate generation and regime selection both matter, and the second is the larger gap.

The missing capability is calibrated regime detection: deciding when *0.61* is a decimal answer, when it is a percent written as a decimal, and when a larger or smaller unit is required by the axis.

7 Conclusion

Scale correction in chart QA is directional and regime-dependent. In answer-tuned Qwen2.5-VL-7B, prefer-large scale-contrastive adaptation recovers most blind-verified strict test-set scale errors, and compute-matched controls attribute the effect to directional supervision rather than extra training. Globally, the same adapter's aggregate effect flips sign with a set's direction composition. Blind sampling shows the model usually already generates the correct scale; verifiers, calculators, and label-free selectors recover only a small, source-specific part of that headroom. For percentage–decimal, factor-of-100 errors on ChartQA-like data, reliable scale-regime selection is the central open problem. Systems should evaluate scale reliability by regime with fix–break accounting, not deploy automatic correction on aggregate gains.

Limitations

Scope of the scale definition. Our strongest evidence concerns factor-of-100 percentage–decimal errors in ChartQA-derived data; 92.8% of image-evaluable ScaleQA-Natural items involve explicit percent evidence. Factors of 10 and 1000, currency units, scientific notation, logarithmic axes, percentage-point differences, and mixed units appear only in small groups and receive no systematic evaluation. Claims about "scale reliability" in general should be read as claims about this regime until the evaluation is broadened.

Small verified slices. The verified strict test subset contains 17 examples, so one example corresponds to roughly six percentage points; we report counts with Wilson intervals and treat recovery as diagnostic-slice evidence, not a population estimate. ScaleQA-Natural is a mined, author-adjudicated set; the inclusion audit reviewed retained positives only and does not bound false negatives in the rejected pool.

Selector evaluation. Selector variants were iterated on the same diagnostic collection with repeated held-out-fold evaluation rather than a pre-registered test partition or nested cross-validation, so the reported advantage over self-check is exposed to researcher-level overfitting; the leave-source-out analysis is our mitigation, and it shows the advantage is fragile. The selector also requires chart-context extraction, auxiliary prompts, and calibration whose latency and cost we have not characterized, so we do not call it deployable.

Models and transfer. The main positive result uses Qwen2.5-VL-7B-Instruct in an answer-tuned setting. InternVL2.5-1B/8B/26B diagnostics and PlotQA, ChartBench, MathVista, and InfographicVQA subsets use differing prompts, subset sizes, and sample definitions; they show that ratio-related errors occur elsewhere, not that prevalence or taxonomy is stable across models. None of these subsets is a full-benchmark or state-of-the-art comparison.

Deployment risk. Incorrect scale interpretation can be consequential in financial, scientific, policy, or medical charts, and the present results do not establish that any tested intervention is safe in those settings. Silently rescaling a numerically plausible answer can convert a visible error into an invisible one. Benchmark accuracy also differs from semantic correctness: *0.61* and *61%* may be equivalent where the unit is explicit and seriously wrong where it is not. We recommend against deploying automatic scale correction policies on the basis of these results.

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A Implementation Details

Appendix roadmap. The main paper compresses the evidence into the story that scale correction is directional and that the hard part is choosing the right scale regime. This appendix gives the details behind that claim. Section B documents how ScaleQA-Natural is mined, audited, and separated from training and official evaluation. Section C reports development evidence showing that the scale-adapter gain is real but narrow. Section D then follows the negative results: stronger prompts, verifiers, table calculators, candidate likelihood scoring, Bayesian fusion, and cross-backbone checks often expose the right answer in the candidate pool, but still fail to choose it reliably. This is why we frame adapters and verifiers as diagnostic probes rather than as a complete repair.

Adapters. The answer-tuned setting continues from the ChartQA/TextVQA answer-only LoRA adapter used in the evaluation pipeline. The answer-only adapters use rank $r = 16$, LoRA $\alpha = 32$, dropout 0.05, and target the language-model projection and MLP modules `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj`. Answer-only training uses AdamW with learning rate 2×10^{-5}. The final Qwen scale adapters use large-positive pairs with $K = 300$, 300 optimization steps, AdamW learning rate 10^{-6}, weight decay 0.01, gradient clipping at 1.0, $\beta = 2.0$, and margin $m = 0.10$. The clean direct setting starts from the raw Qwen2.5-VL-7B-Instruct backbone and applies only the

scale-contrastive update. The InternVL2.5-1B adapter check uses the same pair construction, rank $r = 8$, LoRA $\alpha = 16$, dropout 0.05, 200 optimization steps, and the same language-model target modules.

Scale-pair construction. [Fill in from the pair-mining script: (1) which training rows qualify (single numeric gold answer, nonzero); (2) how the alternative y^- is formed ($\times 100$ or $/100$; rounding to the gold answer's number of decimals; sign preserved; percent sign retained if present in gold); (3) how prefer-large/prefer-small pairs assign y^+; (4) how balanced, all-pair, and real-conflict pair sets are sampled to $K = 300$; (5) treatment of scientific notation and unit tokens; (6) how chart-context strings are extracted for selectors (OCR engine, fields). Also list selector architecture and hyperparameters, candidate-likelihood normalization across prompts, and seeds/item IDs for external subsets.]

Why separate answer-tuned and clean-direct settings. The answer-only adapter is not a scale intervention. It mostly makes the model better aligned to short chart answers, numeric formatting, and the evaluation prompt. Keeping this setting separate matters because the clean-direct scale-only adapter is unsafe on the raw backbone (Table 44), while the answer-tuned setting can absorb a small scale-specific update without the same global collapse. This difference suggests that scale correction is not a standalone numerical operation. It depends on a model already being in a chart-answering regime where answer extraction, formatting, and short-answer calibration are reasonably stable.

Decoding and scoring. All single-adapter systems use the same short-answer prompt and deterministic short-answer decoding template: the question is followed by "Answer briefly." and generation uses greedy decoding with `do_sample=False` and `max_new_tokens=24`. We evaluate numeric answers with the repository evaluator, which uses normalized exact matching for text and a 5% numerical tolerance for numeric answers.

ScaleQA-Natural annotation. Candidate scale conflicts are mined from model predictions and automatically prefilled with numeric-ratio features. The review workflow assigns labels including scale diagnostic candidate, requires image inspection, non-scale, uncertain, and duplicate. Positive labels require that the gold answer and prediction refer to the same chart quantity but differ primarily by a simple scale relation; wrong entity, wrong operation, OCR failure, arbitrary numeric miss, ambiguous question, and duplicate cases are excluded or sent to adjudication. The final ScaleQA-Natural-520 set contains author-adjudicated positive candidates after question/gold deduplication. The image-evaluable subset contains 473 examples with available ChartQA image rows. The audit workflow exports three files: the 739-row candidate audit, a conflict queue, and a 108-row final-positive adjudication queue. After this adjudication, an independent second reviewer audited the 520 final positive inclusions and agreed with all inclusion labels. The release package is intended to include item IDs, source split and row index, question, gold answer, model prediction, subtype, direction, review label, audit metadata, and scripts for reproducing the filters. We reference ChartQA images by source row rather than redistributing them.

Annotation guideline. Annotators mark an item as a scale diagnostic only when the prediction and gold answer are both grounded in the chart question and differ primarily by a simple multiplicative scale relation, such as percentage versus decimal form or a visible unit-scale mismatch. Items are rejected when the prediction reflects the wrong visual entity, wrong operation, OCR failure, ambiguous wording, duplicate evidence, or an arbitrary numeric miss. Ambiguous cases go to a review queue and are not included without adjudication. Annotation was performed from the chart image together with the question and answers, not from metadata alone.

PlotQA external subset. The PlotQA external check uses a 500-example subset sampled from the official PlotQA test parquet after filtering to examples with a single numeric answer. We use the same relaxed numeric evaluator as the ChartQA experiments. This subset is intended as an external numeric-answer transfer check rather than a full PlotQA bench-

mark.

B Data Flow and Annotation Audit

ScaleQA-Natural is useful only if it is not mistaken for an ordinary benchmark split or for hidden training data. Table 8 therefore separates five roles: answer-only tuning, scale-pair training, development analysis, diagnostic set construction, and held-out or external evaluation. The reported Qwen scale adapters are trained from ChartQA training rows and training-time prediction caches; ScaleQA-Natural is used to diagnose regimes and evaluate policy selection, not to train those scale adapters. The held-out ChartQA test result remains the official aggregate task result.

The annotation story has two parts. First, strict factor-of-100 failures are sparse in any single ChartQA output file (Table 10), so a single split underestimates the phenomenon. Second, mining introduces the opposite risk: plausible numeric ratios can be false positives when the model answered the wrong entity or operation. We therefore use ScaleQA-Natural as a diagnostic set whose positives require author adjudication and an independent inclusion audit (Table 13), and we keep uncertain/rejected examples visible (Table 11) rather than silently discarding them. The resulting set is deliberately conservative: it favors precision of the scale-conflict label over treating every ratio-like mismatch as a scale error.

C Additional Results

Tables moved from the main text for space: dataset statistics, the annotation audit, the transfer-boundary summary, the pair-mode ablation, the subtype router, and the ratio-rule test subset.

Table 14 separates answer tuning from scale tuning. The raw pipeline obtains 78.04 on ChartQA test; answer tuning reaches 88.60. Continuing from that adapter, prefer-large scale-contrastive tuning reaches 89.32 over five seeds, +0.72 points with paired bootstrap 95% CI $[+0.38, +1.09]$. Clean scale-only adaptation from the raw backbone hurts test accuracy by 3.5–6.7 points. The aggregate gain is small because strict scale errors are rare (Table 10); non-scale accuracy moves only from 89.35 to 89.66, and TextVQA-500 is unchanged at 94.20 (a small adjacent-task check, not a general safety guarantee).

Table 15 answers the attribution question. On the verified strict test errors the answer-tuned baseline recovers 0/17. Continuing with ordinary answer examples recovers 1/17, and non-directional or random-negative continuation is statistically tied with the baseline on ChartQA validation. Directional scale supervision recovers 12/17 in the three-seed control setting and 14/17 with the main five-seed prefer-large adapter. The intervals are wide because the slice is small, but they exclude the control outcomes: the targeted effect comes from directional scale supervision, not from 300 more optimization steps. We report this as a verified diagnostic-slice result, not as a population-level recovery rate. Per-seed recovery counts and the identities of consistently and inconsistently recovered items are given in Appendix E.

C.1 ScaleQA-Natural

The second question is whether the same intervention is safe when scale conflicts are collected across regimes rather than filtered to one verified recovery subset.

ScaleQA-Natural reveals why a single aggregate number is insufficient. On ScaleQA-Natural-473, answer tuning improves the raw backbone from 45.67 to 50.95, a scale-aware prompt drops to 44.19, and the global prefer-large adapter reaches 49.51, *below* the answer-tuned base. On ScaleQA-Core, balanced and all-pair variants outperform the answer-tuned base; on the InternVL-mined subset, prefer-small adaptation is far stronger than every other global policy (74.90 versus 33.23 balanced). Directionality is real, but it is not "large is good, small is bad": the safe direction depends on the source and subtype of the conflict, and a global adapter that repairs the test-set slice breaks a comparable number of other examples (§6).

Table 17 adds stronger training-free baselines. A self-check prompt reaches 54.55, outperforming the direct prompt and the global adapter; majority voting adds little, while oracle prompt choice reaches 67.44. Global adapters are therefore not the best interven-

Component	Source	Used for	Leakage control
Answer-only tuning	ChartQA/TextVQA training data	Starting adapter for the main Qwen setting	Reported separately from scale tuning; not counted as a scale-objective gain
Scale-pair training	ChartQA training rows and training-set prediction/log-prob caches	LoRA scale adapters and pair-mode ablations	Official ChartQA test and PlotQA are not used for scale-adapter gradients
Development analysis	ChartQA validation and ScaleQA-Core diagnostics	Pair-mode selection, seed checks, and failure analysis	Reported as development evidence; held-out ChartQA test is the main official result
ScaleQA-Natural	Mined model outputs, machine audit, and author adjudication	Diagnostic regime analysis and router experiments	Used as a diagnostic set, not as ordinary task training data for the reported Qwen scale adapters
Held-out/external evaluation	ChartQA test, TextVQA-500, PlotQA numeric-500	Final task result, safety, and external robustness check	Not used to train the reported scale adapters

Table 8: Data flow and split hygiene.

Audit view	Source / flag	Count	Share
Image-evaluable 473	train_shard1	314	66.4%
Image-evaluable 473	val	71	15.0%
Image-evaluable 473	train	47	9.9%
Image-evaluable 473	test	39	8.2%
Image-evaluable 473	train_shard2	1	0.2%
Image-evaluable 473	train_shard3	1	0.2%
Final 520	Contains ChartQA train occurrence	336	64.6%
Final 520	Contains ChartQA validation occurrence	59	11.3%
Final 520	Contains ChartQA test occurrence	23	4.4%

Table 9: ScaleQA-Natural source audit. The set is a diagnostic slice mined from multiple model outputs and source rows, not an ordinary training/test split or leaderboard. Occurrence flags are counted over the final 520 positives; source-split rows are counted over the 473 image-evaluable items.

Model / pipeline	Split	Strict errors	Rate
Qwen2.5-VL-7B + answer/large	ChartQA test	10 / 2500	0.40%
Qwen2.5-VL-3B direct/large	ChartQA test	19 / 2500	0.76%
Qwen2.5-VL-3B direct/large	ChartQA val	21 / 1920	1.09%
InternVL2.5-1B	ChartQA test	3 / 2500	0.12%
InternVL2.5-1B	ChartQA val	17 / 1920	0.89%

Table 10: Strict factor-of-100 scale-error prevalence in available ChartQA validation/test prediction files. These sparse natural subsets motivate the larger ScaleQA-Natural diagnostic set.

tion on this set; the question becomes whether the right answer is missing or merely not selected.

Table 18 addresses a construction concern directly: because ScaleQA-Natural is mined through multiplicative relations and the structured oracle pools contain explicit scale variants, oracles over those pools are high partly by construction. The blind-generation row removes both. With no scale prompt and no transformation, 20 stochastic samples already contain a correct answer for 79.3% of examples, against 52.4% for a single sample. The model frequently generates the correctly scaled answer on its own; deterministic variants raise coverage further to 91.75–92.60, but most of the gap to deployable methods is already present without them.

C.2 Cross-Backbone and External Checks

The final experimental question is scope: does the phenomenon extend beyond the main Qwen setting, and does the intervention transfer with it?

The development results explain why the paper does not present the large-positive adapter as a general correction method. On ChartQA validation, the adapter recovers strict gold-scale errors much more often than it changes ordinary examples (Tables 23 and 25), but the aggregate confidence interval is small and overlaps zero on the full split. This is exactly the diagnostic pattern we care about: the intervention finds a real failure mode, but it is not a broad accuracy engine.

Table 26 is another reason we avoid a "more LoRA pairs solves scale" claim. Gains appear quickly and then saturate, while the subtype table in the main paper shows that the opposite direction requires a different policy. The ablation in Table 27 should therefore be read as a probe-selection experiment: pair direction changes which mistakes are repaired and which safe cases are put at risk.

Audit label	Auto subtype	Question snippet
uncertain	percent_to_decimal	How much Good more than Bad?
uncertain	value_scaled_up	How many diamond cutting and polishing jobs were there in the Far East i...
uncertain	decimal_to_percent	How many respondents say no?
uncertain	decimal_to_percent	How many says that internet has good influence on education?
uncertain	decimal_to_percent	How confident are people about Angela Merkel doing the right thing regar...
uncertain	decimal_to_percent	How many more people answered Very acceptable for the most and least pop...

Table 11: Examples from the negative/uncertain audit queue. These candidates contain a plausible scale relation but are not included as positive diagnostics without stronger evidence that the prediction and gold answer refer to the same chart quantity.

Group	Item	Count
Reviewed diagnostic set	Annotated ScaleQA-Natural items	520
Reviewed diagnostic set	Directly image-evaluable items	473
Reviewed diagnostic set	Core reviewed seed set	403
Reviewed diagnostic set	Image-assisted additions	86
Reviewed diagnostic set	Other self-annotated additions	31
Diagnostic sets	ScaleQA-Natural-520	520
Diagnostic sets	ScaleQA-Natural-520, image-evaluable	473
Diagnostic sets	ScaleQA-Natural-Core	403
Diagnostic sets	InternVL-mined subset	192
Subtypes	Value scaled down	120
Subtypes	Decimal to percent	114
Subtypes	Value scaled up	104
Subtypes	Percent to decimal	96
Subtypes	Minor factor or unit candidate	42
Subtypes	Minor factor percentage candidate	28
Subtypes	Untyped or mixed factor candidate	16

Table 12: ScaleQA-Natural construction statistics. It is a mined diagnostic set; official ChartQA validation/test remain the task evaluations.

Audit item	Count/Score	Notes
Candidate pool	739	First-pass reviewed candidates
Observed agreement	0.633	Three-way labels
Cohen's κ	0.443	Include/review/reject
Conflicts for review	271	Exported separately
Final positives	520	Author-adjudicated
Second-pass include	412	Deterministic audit
Second-pass review/reject	108	Manually adjudicated
Independent inclusion audit	520	Second human reviewer
Second-review agreement	1.000	Final positives; zero label changes

Table 13: ScaleQA-Natural audit stages. Cohen's κ compares the deterministic metadata audit with first-pass labels; the second human reviewer audited final inclusions only.

D Additional Transfer and Analysis Tables

This section contains the main negative evidence behind the paper's conclusion. If scale correction were a simple numeric normalization problem, stronger backbones, larger candidate pools, chart-to-table calculators, or Bayesian policy fusion should close the gap. Instead, the same pattern recurs. The correct answer is often present under some candidate or policy oracle, but deployable selectors remain directionally biased. This supports the paper's central distinction: the phenomenon transfers more robustly than the intervention.

Tables 28–41 show the transfer boundary.

System	Starting point	Test acc.	Std.	Delta
Raw Qwen2.5-VL-7B	Raw backbone	0.7804	–	–
Answer-tuned LoRA	Answer-only	0.8860	–	+0.1056
Answer-tuned + large-positive scale	Answer-tuned	0.8932	0.0003	+0.0072
Clean direct large-positive scale	Raw backbone	0.7456	–	-0.0348
Clean direct small-positive scale	Raw backbone	0.7136	–	-0.0668

Table 14: Held-out ChartQA test attribution for Qwen2.5-VL-7B. The scale update is measured on top of the answer-tuned adapter; clean direct scale-only tuning is a safety control. "Large-positive" denotes the prefer-large adapter.

Continuation from answer-tuned adapter (300 steps, 3 seeds)	ChartQA val. acc.	Verified strict test errors recovered	
None (answer-tuned baseline)	0.8640	0/17	[0.000, 0.184]
Ordinary answer replay	≈0.8640	1/17	[0.010, 0.270]
Random / shuffled numeric negatives	≈0.8640	[fill]	
Non-directional balanced pairs	≈0.8640	[fill]	
Directional scale pairs (prefer-large)	0.8695	12/17	[0.469, 0.867]
Prefer-large adapter, main setting (5 seeds)	–	14/17	[0.590, 0.938]

Table 15: Compute-matched controls. Every row continues the same answer-tuned adapter for the same number of steps; only the continuation data differs. Recovery is reported as counts over the 17 blind-verified strict test errors with Wilson 95% intervals.

Larger InternVL models improve the base level on ScaleQA-Natural and external numeric subsets, but the same large/small/balanced scale adapters do not become uniformly helpful. On ScaleQA-stress1000, for example, InternVL2.5-8B adapters slightly reduce accuracy relative to the base model; on MathVista and InfographicVQA the changes are small; on PlotQA the adapter can hurt. These results do not refute the existence of scale inconsistency. They show that a global scale intervention learned in one model/task setting is not the right abstraction for deployment.

The selector and verifier results give the sharpest version of the bottleneck. Self-check prompting is already a strong fixed policy, and the best no-leak selector improves on it only modestly (Table 34). Yet the candidate and policy oracles are much higher. The gap is not caused by a failure to generate alternatives: it is caused by failures to rank alternatives under chart-specific scale evidence. The structured

Set	N	Raw	Ans.	Prompt	Large	Bal.	All	Real	Small
ScaleQA-Natural-520	473	0.4567	0.5095	0.4419	0.4951	–	–	–	–
ScaleQA-Core	403	0.4591	0.4963	0.4293	0.4844	0.5092	0.5047	0.5002	0.4174
InternVL-mined	192	0.1510	0.3281	0.0990	0.1896	0.3323	0.2771	0.2510	0.7490

Table 16: ScaleQA-Natural diagnostic results. The first row is the image-evaluable ScaleQA-Natural-473 subset; ScaleQA-Core and InternVL-mined report five-seed pair-mode means. Large/Small denote prefer-large/prefer-small adapters.

Prompting strategy	Accuracy
Direct answer	0.5243
Axis-aware prompt	0.5307
Math-check prompt	0.5137
Scale prompt	0.4905
Self-check prompt	0.5455
Unit-guard prompt	0.5180
Majority vote	0.5264
Oracle prompt choice	0.6744

Table 17: Training-free prompting baselines on ScaleQA-Natural-473 with the answer-tuned Qwen2.5-VL-7B pipeline.

Candidate source on ScaleQA-Natural-473	Accuracy / coverage
Single greedy sample (answer-tuned)	0.524
Blind pass@20, no scale prompt, no transforms	0.793
Prompt + adapter candidates (policy-pool oracle)	0.9004
Top-8 candidates incl. deterministic scale variants	0.9175
Augmented structured oracle	0.9260

Table 18: Candidate coverage on ScaleQA-Natural-473 decomposed by candidate source. Blind sampling uses no scale prompt and no deterministic transformation, so its coverage is not implied by the mining rule.

Check	Result	Takeaway
InternVL2.5-8B ScaleQA	0.4672 → 0.4723	tiny adapter gain
InternVL2.5-26B external	0.4080–0.7200	stronger diagnostic backbone
Qwen7 external adapters	no change	no broad transfer win
ScaleQA-stress1000	all 0.6940	global policies tie

Table 19: Transfer boundary summary. Full cross-backbone and external rows are in the appendix.

Method	Accuracy	Delta
Large-positive	0.863500	0.003100
Real conflicts	0.862500	0.002100
Balanced	0.860900	0.000500
All pairs	0.860900	0.000500
Base	0.860400	0.000000
Small-positive	0.845300	-0.015100

Table 20: Pair-mode ablation on ChartQA validation. Prefer-small (small-positive) adaptation is locally useful in some regimes but globally unsafe on this split.

verifier makes this especially clear. It is strong when the prediction is too small but nearly collapses when the prediction is too large (Table 37), even though the augmented oracle is high in both directions.

D.1 Selector Variant Ladder

The main text reports only the strongest label-free selector. Table 34 lists every variant for completeness. v8–v17 were developed sequentially on the same diagnostic collection using repeated held-out-fold evaluation (not nested cross-validation); variants after v10b preserve its score but do not improve it. We retain the ladder so that the extent of iteration, and hence the exposure to researcher-level overfitting, is visible.

The table-calculator baseline is a useful stress test because it is closer to symbolic reasoning than to answer-style prompting. It performs well on percent-to-decimal and value-scaled-down cases, but almost never solves decimal-to-percent or value-scaled-up cases (Table 38). That asymmetry mirrors the adapter story: changing the evidence source does not remove the need to identify the regime first.

The subtype and taxonomy tables explain why the task is hard for deployable selectors. Most image-evaluable examples contain explicit percent evidence, but the correct action is not always "add a percent" or "multiply by 100". In decimal-to-percent and value-scaled-up cases, the small-positive policy is often best; in percent-to-decimal and value-scaled-down cases, the large-positive policy is best (Table 42). The same words—percent, average, difference, value lookup—appear across regimes (Table 43), so a selector needs to reason over the chart, the question, the base answer, and the candidate answer jointly rather than keying off a single unit token.

Finally, Table 44 is the safety counterexample. A scale-only update from the raw Qwen backbone degrades ChartQA valida-

Set	Fixed best	Pred. router	Router acc.	Oracle subtype	Instance oracle
ScaleQA-Core	0.4997	0.5071	0.5444	0.8077	0.8081
InternVL-mined	0.7292	0.7292	0.6918	0.7741	0.7792

Table 21: Subtype routing on ScaleQA-Natural diagnostics. Fixed best is the best single global policy; oracle columns use gold information and are upper bounds.

Split	Base	Ours	Std.	Delta	N
All test examples	0.8860	0.8932	0.0003	+0.0072	2500
Strict gold-scale errors	0.0000	0.4952	0.0543	+0.4952	21
Non-gold-scale examples	0.8935	0.8966	0.0005	+0.0031	2479

Table 22: Held-out ChartQA test subset analysis with the ratio-rule strict subset (21 candidates before blinded verification). The strict subset is baseline-zero by construction.

tion, ChartQA test, and ScaleQA-Core, even though it helps a separate InternVL-mined subset in the opposite direction. This is why the paper treats scale adapters as probes of directional asymmetry, not as a universal repair. The case studies in Table 45 give the qualitative version of the same lesson: the model can repair 0.61→61 and still over-correct a true small decimal such as 0.0593.

Overall appendix takeaway. Taken together, the appendix strengthens the main claim in a deliberately conservative way. Scale-like mistakes are not frequent enough in a single benchmark split to support sweeping accuracy claims, but they are systematic enough to reveal a reliability failure once mined and audited. Global adapters expose the directionality, training-free self-checks provide a strong practical baseline, and expanded candidate pools show that many failures are recoverable in principle. The unresolved problem is selecting the right regime without gold subtype labels. This is the reason the paper argues for regime-aware scale reliability rather than presenting unconditional scale correction as solved.

E Verified Strict Test Errors and Overlap Check

Protocol. All ChartQA test examples with a single numeric gold answer were scanned with the ratio rule of §3. Each strict candidate was shown to the annotator in a blinded interface containing only the chart image, question, gold answer, and the answer-tuned model's prediction; the candidate group, numeric ratio, and relation label were hidden. The annotator assigned one of: *true scale error*, *true scale error with confound* (e.g., a format convention also at play), or *not a scale error*. Thirty randomly sampled non-candidate test errors were labeled under the same interface as a false-negative check.

Outcome. Of 18 candidates, 17 were labeled true scale errors and 16 clean true scale errors; 1 of 30 non-candidates was judged a missed scale error (Wilson 95% intervals: $[0.742, 0.990]$, $[0.672, 0.969]$, and $[0.006, 0.167]$). [Reconcile with the 21 strict candidates in Table 22: state whether the 18 result from a stricter parse (e.g., sign-preserving, percent-aware) and list the 3 excluded items.]

Per-item table. [Insert table: item ID, chart source row, question, gold, baseline prediction, prefer-large prediction per seed, control predictions, manual label, recovered-by-how-many-seeds.]

Row-level overlap. Source split of the 473 image-evaluable items: train shard 1, 314; validation, 71; train, 47; test, 39; train shards 2–3, 2. Items from any training split (363) are the training-row subset of Table 4. The analysis is in `results/scaleqa_natural/row_overlap_split_analysis.c` the leave-source-out run is `scripts/experiment_policy_selector_v10b_leave_source` with outputs under `results/scaleqa_natural/leave_source_out_v10b/`

F Oracle Definitions

Candidate oracle Accuracy if the best candidate in a given pool is always selected.

Policy-pool oracle Candidate oracle over the pool formed by one answer per policy (base, prompts, adapters).

Instance oracle Policy-pool oracle evaluated per instance with gold subtype known.

Augmented structured oracle Candidate oracle after adding deterministic ×100 and /100 variants of each pooled candidate.

Blind pass@k Fraction of examples for which at least one of k stochastic samples from the answer-tuned model (no scale prompt, no transform) is correct.

Split	N	Base	Ours	Delta	CI low	CI high	$p(\Delta \leq 0)$
All examples	1920	0.860420	0.862500	0.002080	-0.001250	0.005520	0.113500
Numeric-answer examples	1452	0.860880	0.862810	0.001930	-0.002200	0.006060	0.187800
Scale-keyword numeric examples	1179	0.845630	0.847840	0.002210	-0.002880	0.007460	0.202500
Strict gold-scale errors	23	0.000000	0.113040	0.113040	0.000000	0.252170	0.036800
Non-gold-scale examples	1897	0.870850	0.871590	0.000740	-0.002210	0.003800	0.327100

Table 23: Bootstrap analysis on ChartQA validation for one large-positive adapter snapshot. The full-set confidence interval overlaps zero, so we emphasize the directionality and diagnostic analyses rather than overclaiming aggregate accuracy.

Split	Base	Ours	Std	Delta	N
All examples	0.860400	0.864000	0.000600	0.003500	1920
Strict gold-scale errors	0.000000	0.208700	0.036400	0.208700	23
Non-gold-scale examples	0.870800	0.871900	0.000400	0.001100	1897

Table 24: ChartQA validation subset analysis for the answer-tuned large-positive scale adapter. This was used as development evidence; the main official result is the held-out test result in Table 22.

Split	N	Base	Mean fixes	Mean breaks	Any fix	All fix	Any break	All break
All examples	1920	0.860400	0.021400	0.010900	0.005700	0.003100	0.002600	0.001600
Base-wrong examples	268	0.000000	0.153000	0.000000	0.041000	0.022400	0.000000	0.000000
Base-correct examples	1652	1.000000	0.000000	0.012700	0.000000	0.000000	0.003000	0.001800
Strict gold-scale errors	23	0.000000	0.565200	0.000000	0.130400	0.087000	0.000000	0.000000
Non-gold-scale examples	1897	0.870800	0.014800	0.011100	0.004200	0.002100	0.002600	0.001600

Table 25: Seed-level fix and break rates on ChartQA validation.

Method	Pairs per kind	Accuracy	Delta
Base	0	0.850700	0.000000
Large-25	25	0.856000	0.005300
Large-50	50	0.856000	0.005300
Large-100	100	0.855300	0.004700
Large-200	200	0.856000	0.005300
Large-300	300	0.855300	0.004700
Large-400	400	0.855300	0.004700
Large-600	600	0.856000	0.005300
Large-800	800	0.856000	0.005300

Table 26: Large-positive pair-size curve on the ChartQA-1500 setting. Gains appear with few pairs and then saturate.

Method	Overall	Δ	Gold-scale	Δ	Non-scale	Δ
Base	79.00	+0.00	0.00	+0.00	82.81	+0.00
Scale-aware prompt	79.00	+0.00	0.00	+0.00	82.81	+0.00
Direct SFT (800)	78.20	-0.80	4.35	+4.35	81.76	-1.05
Direct SFT (1500)	77.40	-1.60	4.35	+4.35	80.92	-1.89
Mixed contrastive	79.40	+0.40	8.70	+8.70	82.81	+0.00
Balanced contrastive	79.20	+0.20	4.35	+4.35	82.81	+0.00
Large-only contrastive	79.00	+0.00	4.35	+4.35	82.60	-0.21
Real contrastive	79.40	+0.40	8.70	+8.70	82.81	+0.00
Real contrastive (5 seeds)	**79.52**	**+0.52**	11.30	+11.30	82.81	+0.00

Table 27: Ablation results on ChartQA-500. Gold-scale denotes strict 100× scale-error cases where the base model predicts an answer at the wrong numerical scale.

Backbone	Split	N	Accuracy	Strict scale errors
Qwen2.5-VL-7B	ChartQA test	2500	0.7804	10
InternVL2.5-1B	ChartQA validation	1920	0.7745	17
InternVL2.5-1B	ChartQA test	2500	0.7524	3
InternVL2.5-1B	ScaleQA-Natural-520	473	0.3340	–

Table 28: Cross-backbone strict scale-like error diagnostic.

Set	Model	Base	Large	Small	Bal.
ScaleQA-Natural-520	InternVL2.5-1B	0.3340	0.3446	0.3446	–
ScaleQA-Natural-520	InternVL2.5-8B	0.4672	0.4685	0.4723	0.4689
ScaleQA-Natural-520	InternVL2.5-26B	0.4989	–	–	–
ScaleQA-stress1000	InternVL2.5-8B	0.7320	0.7272	0.7276	0.7256
ScaleQA-stress1000	Qwen2.5-VL-7B	0.6940	0.6940	0.6940	0.6940
PlotQA-557	InternVL2.5-8B	0.6912	0.6829	0.6862	0.6840
PlotQA-500	InternVL2.5-26B	0.7200	–	–	–
ChartBench-500	Qwen2.5-VL-7B	0.4820	0.4820	0.4820	0.4820
ChartBench-500	InternVL2.5-8B	0.3520	0.3512	0.3516	0.3520
ChartBench-500	InternVL2.5-26B	0.4080	–	–	–
MathVista-243	Qwen2.5-VL-7B	0.6708	0.6708	0.6708	0.6708
MathVista-243	InternVL2.5-8B	0.6337	0.6379	0.6387	0.6387
MathVista numeric	InternVL2.5-26B	0.6749	–	–	–
InfographicVQA-300	Qwen2.5-VL-7B	0.7767	0.7767	0.7767	0.7767
InfographicVQA-300	InternVL2.5-8B	0.6067	0.6087	0.6073	0.6093
InfographicVQA-300	InternVL2.5-26B	0.6300	–	–	–

Table 29: Full cross-backbone and external numeric transfer table.

Backbone and setting	N	Accuracy	Std.
InternVL2.5-1B base	473	0.3340	–
InternVL2.5-1B large-positive LoRA	473	0.3446	–
InternVL2.5-1B small-positive LoRA	473	0.3446	–
InternVL2.5-8B base	473	0.4672	–
InternVL2.5-8B large-positive LoRA	473	0.4685	0.0017
InternVL2.5-8B small-positive LoRA	473	0.4723	0.0035
InternVL2.5-8B balanced LoRA	473	0.4689	0.0025

Table 30: InternVL scale adaptation on ScaleQA-Natural-520.

InternVL2.5-8B setting	N	Accuracy	Std.
Base	1000	0.7320	–
Large-positive adapter	1000	0.7272	0.0016
Small-positive adapter	1000	0.7276	0.0034
Balanced adapter	1000	0.7256	0.0023

Table 31: InternVL2.5-8B on a synthetic ScaleQA-stress1000 set.

Prompting strategy	Accuracy
Direct answer	0.5243
Axis-aware prompt	0.5307
Math-check prompt	0.5137
Scale prompt	0.4905
Self-check prompt	0.5455
Unit-guard prompt	0.5180
Majority vote	0.5264
Oracle prompt choice	0.6744

Table 32: Training-free prompting baselines on ScaleQA-Natural-520.

Method	Overall	Pred. too large	Pred. too small
Candidate majority	0.5159	0.3029	0.7371
Free-form verifier	0.4588	0.1411	0.7888
Letter verifier	0.4651	0.1411	0.8017
Scale-guard letter verifier	0.4609	0.1452	0.7888
Candidate oracle	0.8288	0.7593	0.9009

Table 33: Multi-candidate verifier on ScaleQA-Natural-520. Candidate oracle selects the best answer from the same candidate pool and is an upper bound.

Baseline or selector	Accuracy	Notes
Self-check prompt	0.5514	Best fixed prompt on held-out splits
v8 no-leak selector	0.5432 ± 0.0501	Question, answer, numeric, and chart-context features
v9 expanded selector	0.5501 ± 0.0460	Adds structured/pairwise policies but does not improve
v10b no-leak selector	0.5667 ± 0.0561	Best clean selector; adds corrected TTA-v2 evidence
v11 log-prob selector	0.5667 ± 0.0561	Log-prob evidence preserves but does not improve v10b
v13 rationale selector	0.5667 ± 0.0561	Rationale-verifier evidence preserves v10b
v14/v15 candidate-likelihood selectors	0.5667 ± 0.0561	Answer/large/base likelihood evidence preserves v10b
v16/v17 top-8 selectors	0.5667 ± 0.0561	Larger candidate pools preserve v10b but raise only oracle
Statistical policy fusion	0.5309 ± 0.0470	Beta/Wilson reliability fusion
Bucketed Bayesian selector	0.5239 ± 0.0480	Bucketed policy priors; over-switches
Bayesian-MLP selector	0.5218 ± 0.0540	Bucket priors plus hidden-layer selector; over-switches
Pairwise top-8 verifier	0.5433	Candidate comparison prompt
Candidate likelihood top-8	0.4503	Log-probability candidate scoring
Structured-context verifier	0.4841	Image, chart context, and top-4 candidates
Structured scale-audit verifier	0.4376	Explicit unit/scale audit prompt
Table extraction + calculator	0.4228	DePlot-style chart-to-table evidence, then symbolic calculation
Original policy-pool oracle	0.7023 ± 0.0384	Upper bound over early policy pool
Expanded policy-pool oracle	0.9004	Upper bound after adding prompts/verifiers
Candidate top-8 oracle	0.9175	Correct answer often appears in candidate set
Augmented structured oracle	0.9260	Upper bound after adding scaled candidate variants

Table 34: Deployable selectors and strong baselines on ScaleQA-Natural. Learned and rule-based selectors use held-out splits, while fixed prompting, verifier, and table-calculator rows are full-set scores. The gap to the oracle rows shows that policy selection remains the bottleneck.

Quantity	Mean	Std.	Additional statistic
No-leak selector	0.5667	0.0561	50 held-out splits
Self-check prompt	0.5514	0.0430	same splits
Selector - self-check	0.0154	0.0492	95% CI [0.0017, 0.0286]
Best fixed policy	0.5714	0.0362	split oracle over fixed policies
Policy oracle	0.9004	0.0238	upper bound
Switch rate	48.9%	39.3%	not using self-check

Table 35: No-leak selector stability on ScaleQA-Natural held-out splits. The improvement over self-check is modest but positive under a split-level bootstrap; the oracle gap remains much larger.

Group	N	Selector	Ans.	Large	Self-check	Best fixed
Direction: prediction too large	241	0.4860	0.2946	0.1452	0.4357	0.4357
Direction: prediction too small	232	0.6534	0.7328	0.8621	0.6595	0.8621
Source: test	39	0.4275	0.3590	0.5385	0.3846	0.5385
Source: val	71	0.6250	0.5915	0.6338	0.6479	0.6479
Source: train	47	0.6303	0.6170	0.5319	0.6809	0.6809
Source: train shard1	314	0.5587	0.4904	0.4522	0.5191	0.5191
Subtype: decimal to percent	109	0.5193	0.2752	0.1468	0.3853	0.3853
Subtype: percent to decimal	74	0.6411	0.8243	0.9324	0.7297	0.9324
Subtype: value scaled down	118	0.6578	0.7034	0.8729	0.6102	0.8729
Subtype: value scaled up	102	0.4098	0.2549	0.0784	0.4216	0.4216

Table 36: Selector and fixed-policy behavior by source, direction, and subtype. "Selector" is the v10b held-out selector score averaged over test-fold appearances. Fixed-policy columns are item-level diagnostics from the candidate pool and are included to show directional reversal, not as a new benchmark leaderboard.

Method	Overall	Pred. too large	Pred. too small
Structured-context verifier	0.4503	0.0954	0.8190
Structured scale-audit verifier	0.4376	0.0996	0.7888
Augmented structured oracle	0.9260	0.9419	0.9095

Table 37: Structured verifier by scale direction. Adding chart context and scaled candidate variants makes the oracle high, but deployable verifiers remain directionally biased.

Subset	Reason	Calculator	Either
All examples	0.3890	0.4080	0.4228
Percent-to-decimal	0.8378	0.8243	0.8378
Value scaled down	0.6949	0.8220	0.8475
Decimal-to-percent	0.0275	0.0183	0.0183
Value scaled up	0.0098	0.0098	0.0098

Table 38: DePlot-style table-extraction plus calculator baseline by subtype on ScaleQA-Natural. The method is strong for some regimes but fails on the opposite direction, again showing that scale correction is regime-dependent.

Policy pool	Best fixed	Learned	+Chart ctx.	Subtype oracle	Inst. oracle
Prompt regimes (473)	0.5408	0.5366	0.5358	0.6192	0.6701
Adapters+prompts (403)	0.5032	0.4467	0.4615	0.8010	0.8410

Table 39: Learned policy routing with question/base-answer numeric features and optional chart-side context extracted from the image. Values are 50-seed means.

Backbone	Method	Accuracy
Qwen2.5-VL-7B	Answer-tuned	0.902
Qwen2.5-VL-7B	Answer-tuned + large-positive scale	0.898
Qwen2.5-VL-7B	Base + scale prompt	0.832
Qwen2.5-VL-7B	TTA majority	0.892
Qwen2.5-VL-7B	TTA oracle prompt choice	0.904
InternVL2.5-1B	Base	0.600

Table 40: External numeric PlotQA-500 check.

Dataset	N	Base	Large	Small	Balanced
PlotQA	557	0.6912	0.6829	0.6862	0.6840
ChartBench	500	0.3520	0.3512	0.3516	0.3520
MathVista	243	0.6337	0.6379	0.6387	0.6387
InfographicVQA	300	0.6067	0.6087	0.6073	0.6093

Table 41: External numeric checks with InternVL2.5-8B.

Subtype	N	Base	Large-pos.	Small-pos.	Best direction
Decimal to percent	109	0.0917	0.1560	0.7890	Small
Value scaled up	102	0.0784	0.0784	0.7059	Small
Percent to decimal	74	0.9054	0.9324	0.0270	Large
Value scaled down	118	0.8475	0.8559	0.0593	Large

Table 42: ScaleQA-Natural subtype analysis on the 403-example reviewed set.

Group	Count	Share
Question pattern		
Percent / percentage	133	28.1
Average / mean	75	15.9
Value lookup	75	15.9
Difference / comparison	65	13.7
Maximum / highest	44	9.3
Minimum / lowest	30	6.3
Unit evidence		
Percent in axis or question	439	92.8
No explicit unit	14	3.0
Large-unit word	13	2.7
Ratio or rate	6	1.3
Currency	1	0.2

Table 43: Fine-grained taxonomy of the image-evaluable ScaleQA-Natural subset. Counts summarize question patterns and unit evidence; categories are not mutually exclusive across the two blocks.

Evaluation set	N	Base	Direct-large	Delta	Direct-small	Delta
ChartQA validation	1920	0.7536	0.7219	-0.0318	0.6948	-0.0589
ChartQA test	2500	0.7804	0.7456	-0.0348	0.7136	-0.0668
ScaleQA-Core	403	0.4591	0.4367	-0.0223	0.4243	-0.0347
InternVL-mined	192	0.1510	0.0938	-0.0573	0.3542	+0.2031

Table 44: Clean direct scale-only adaptation from the raw Qwen2.5-VL-7B backbone.

Type	Question	Gold	Base	Calibrated
Recovered	How much times Madagascar greater than Central Europe data?	32	0.032	32
Recovered	What the difference in value between Asia and Caribbean?	1.18	0.18	1.18
Recovered	What is the highest value of gray bar?	61	0.61	61
Recovered	What is the difference in % between Poland and Australia for people who are worried about losing their job?	17.05	0.1705	17.05
Over-correction	Find out the average of the bottom three countries ??	0.0593	0.06	6.03333

Table 45: Representative recoveries and an over-correction failure.